



Department of Artificial Intelligence & Machine Learning
Academic Year 2026-27

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Date: 13th August 2026.

AIM: To implement Averaging and Median filtering technique on given noisy images (Spatial Domain).

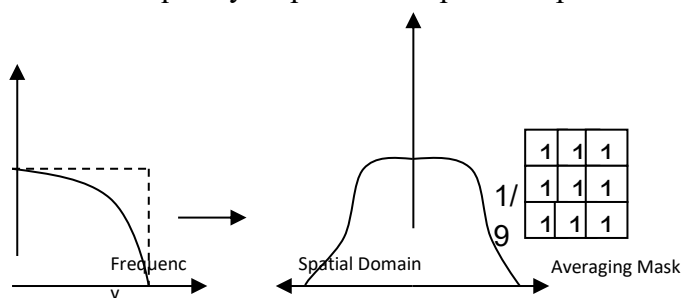
THEORY:

1. Smoothing

Low pass filtering removes the high frequency content from the image. It is used to remove the noise present in the image which is generally a high frequency signal.

2. Averaging Filter

If an image has Gaussian noise, a low pass averaging filter is used to remove the noise. The frequency response and spatial response is show below.



From spatial response we generate the mask that would give us the low pass filtering operation. Here all the coefficients are positive and the multiplying factor for a $M \times N$ matrix is $1 / (M \times N)$.

3. Median Filtering

The averaging filter removes the noise by filtering it till it is no longer seen. But in the process it also blurs the edges. If we use averaging filter to remove salt and pepper noise from an image it will blur the noise but will also ruin the edges. Hence when we need to remove salt and pepper noise we use non-linear filter known as Median Filter. They are also called order statistical filters because their response is based on the ordering and ranking of the pixels contained within the mask.



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Steps to perform median filtering:

1. Assume 3 X 3 empty mask.
2. Place it on left hand side corner.
3. Arrange the 9 pixels in ascending or descending order
4. Choose the median from these nine values.
5. Place this median at the centre.
6. Move the mask in similar fashion as the averaging mask

In median filtering the grey level of the centre pixel is replaced by the median filter value of the neighbourhood



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Code & Output:

```
[ ] import cv2
import numpy as np
from matplotlib import pyplot as plt

img = cv2.imread(r"C:\Users\djsce.student\Desktop\pcb.jfif")
img2 = cv2.imread(r"C:\Users\djsce.student\Desktop\lady.png")

if img is not None:
    img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    plt.imshow(img); plt.imshow(img2)
    plt.title("Noisy Image")
    plt.axis("off")
    plt.show()
else:
    print("Error")
```

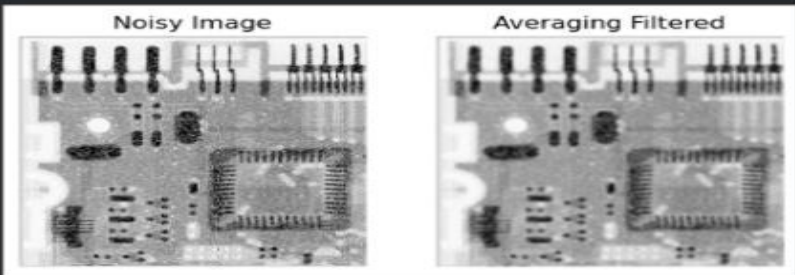
↓



```
[ ] mean_filtered = cv2.blur(img, (3, 3))

plt.subplot(1, 2, 1); plt.imshow(img, cmap='gray'); plt.title('Noisy Image'); plt.axis('off')
plt.subplot(1, 2, 2); plt.imshow(mean_filtered, cmap='gray'); plt.title('Averaging Filtered')
plt.show()
```

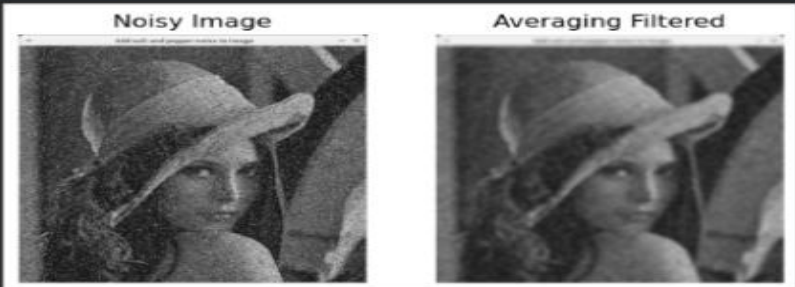
...



```
[ ] mean_filtered = cv2.blur(img2, (10, 10))

plt.subplot(1, 2, 1); plt.imshow(img2, cmap='gray'); plt.title('Noisy Image'); plt.axis('off')
plt.subplot(1, 2, 2); plt.imshow(mean_filtered, cmap='gray'); plt.title('Averaging Filtered')
plt.show()
```

↓





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```
[ ] median_filtered = cv2.medianBlur(img2, 3)
plt.subplot(1, 2, 1); plt.imshow(img2, cmap='gray'); plt.title('Noisy Image'); plt.axis('off')
plt.subplot(1, 2, 2); plt.imshow(median_filtered, cmap='gray'); plt.title('Median Filtered')
plt.show()
```

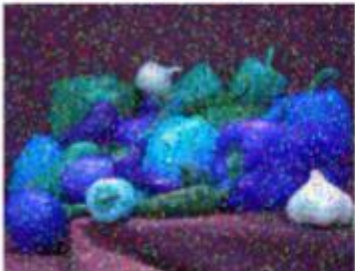
▼



```
[ ] img3 = cv2.imread(r"C:\Users\djsce.student\Desktop\colored.jfif")
```

```
[ ] median_filtered = cv2.medianBlur(img3, 3)
plt.subplot(1, 2, 1); plt.imshow(img3, cmap='gray'); plt.title('Noisy Image'); plt.axis('off')
plt.subplot(1, 2, 2); plt.imshow(median_filtered, cmap='gray'); plt.title('Median Filtered')
plt.show()
```

▼





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Post Lab Session

Q1. Differentiate between the Mean Filter and Median Filter based on the following parameters:

- * Type of filter
- * Operation performed
- * Noise removed
- * Edge preservation
- * Computational complexity

Q. 2 Explain the working principle of the Median Filter. Why is it more effective than the Mean Filter for removing salt-and-pepper noise? Illustrate with a suitable example.

Q. 3 What is the purpose of using image smoothing filters in Computer Vision? Discuss the advantages, limitations, and applications of the Mean Filter and Median Filter.

Q. 4

I =

25	30	28
35	255	32
27	29	31

- a) Compute the output of the Mean Filter for the center pixel.
- b) Compute the output of the Median Filter for the center pixel.
- c) Which filter gives a better result for this image? Justify your answer.



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1/26 EXP-03

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Mean Filter	Median Filter
Linear filter	Non-linear filter
Replaces the center pixel with avg. of neighboring pixels.	Replaces the center with the median of neighboring values.
Effective for Gaussian noise	Best for Salt-and-pepper noise.
Poor, tends to blur edges	Good, preserves edges while removing noise.

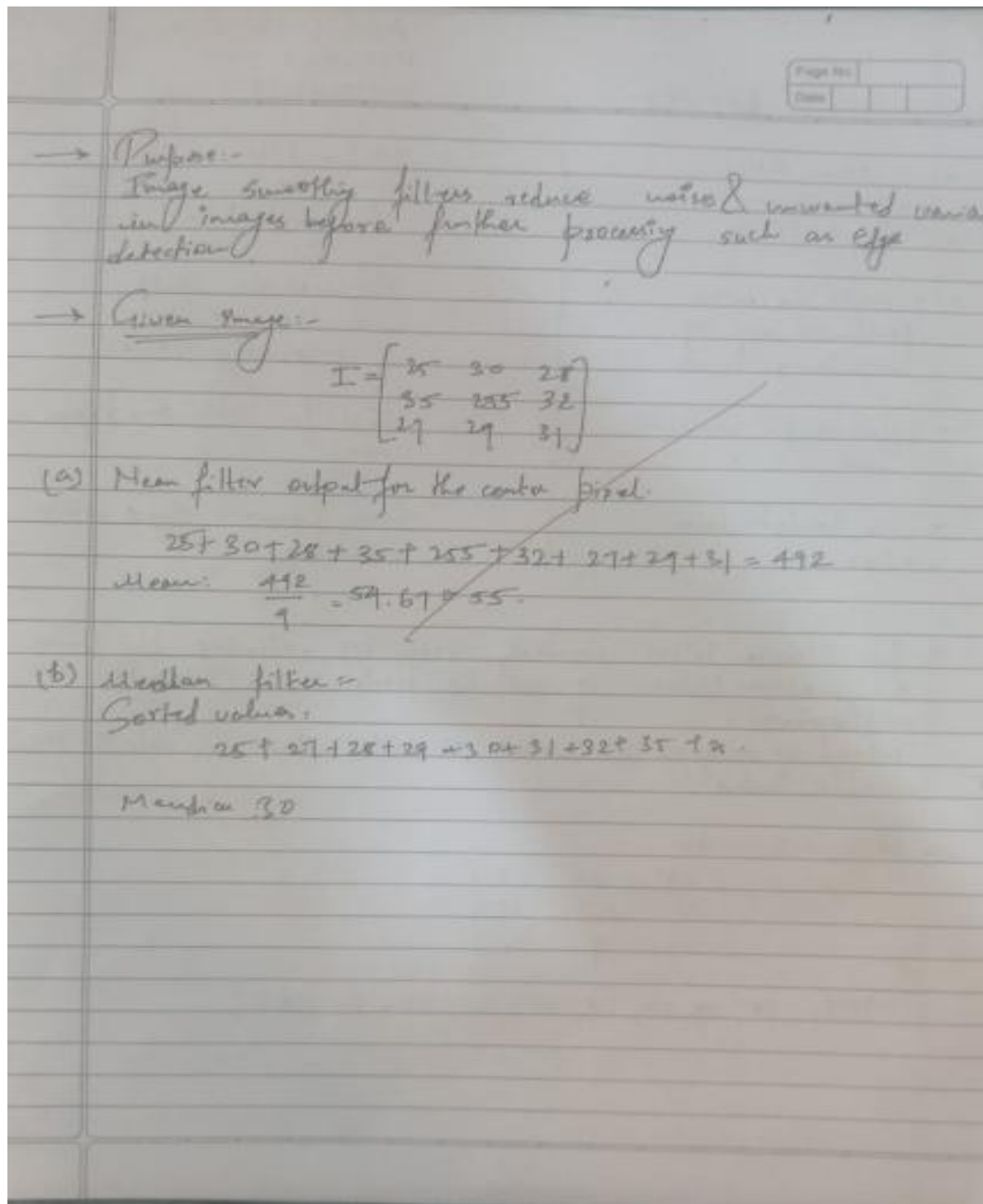
→ The Median Filter removes noise by replacing the center the median value of all the pixels in a selected neighborhood.
Instead of averaging the values, it sorts them & selects the middle value.

→ Eg:- 3x3 window

25	30	28
35	255	32
27	29	31

Sorted: 25, 27, 28, 29, 30, 31, 32, 35, 255.

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Conclusion:

The averaging and median filtering techniques were successfully implemented on noisy images in the spatial domain. The averaging filter effectively reduced Gaussian noise but caused some blurring of edges. The median filter was more effective in removing salt-and-pepper noise while preserving the important edges of the image. Thus, the experiment demonstrated that the choice of filtering technique depends on the type of noise present in the image.



Shri Vile Parle Kelavani Mandal's

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(Autonomous College Affiliated to the University of Mumbai)

NAAC Accredited with "A" Grade (CGPA : 3.18)



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REFERENCE: Digital Image Processing by Gonzalez and Woods, 2nd edition.